



Sustainable synthesis of bimetallic (Zn/Bi)-oxy-chloride-based nanosheets enmeshed rods nanocomposite using bee pollen (ZBCBR) for photo-responsive biocidal agent

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ABSTRACT

Technological advancement in the healthcare sector, especially the discovery of medicines, has positive impacts on human health and the environment. The inappropriate use of medicine causes severe human health issues, mainly the development of antibiotic resistance, thereby necessitating the development of sustainable photo-responsive biocidal agents. The present study describes the synthesis of bimetallic (Zn/Bi)-oxy-chloride-based nanosheets enmeshed rods nanocomposite by using bee pollen (ZBCBR) for photo-responsive biocidal agents. The data suggested that the band gap value of the prepared ZBCBR-based photo-responsive biocidal agents initially increased from ~2.65 eV to ~2.95 eV after incorporation of Zn-metals (0.5 g), whereas further increments of the Zn-metals (0.5–2 g) narrowed the band gap value from ~2.58 eV to ~2.42 eV. Interestingly, upon increasing the Zn-metals, surface charges become positive, which might be beneficial for the interaction with bacterial strains. The synthesized ZBCBR-based photo-responsive biocidal agents were tested against *Escherichia coli* (*E. coli*) and *Staphylococcus aureus* (*S. aureus*) bacterial strains under both dark and solar light irradiation conditions. The ZBCBR-based photo-responsive biocidal agent inhibited/killed both bacterial strains within 24 h. In contrast, ZBCBR-2-based photo-responsive biocidal agent inhibited ~99.99% of bacterial strains within 5 min of solar irradiation at a 2 mg/mL dose. Therefore, the synthesized ZBCBR-based photo-responsive biocidal agent is a simple, sustainable, and efficient antibiotic material.

1. Introduction

Presently, technological development in the healthcare sector, especially the discovery of medicines, has positive impacts on human health and the environment. The inappropriate use of medicines causes severe human health issues, mainly the development of microbial/antibiotic resistance. Although, World Health Organization (WHO) addresses the global concern on the development of antibiotic resistance and issues a guideline for the use of antibiotic drugs. Usually, the emergence of bacterial resistance is mainly due to the inappropriate/excessive use of antibiotic drugs, which convert into multidrug resistance (MDR) and extensive drug resistance (XDR), causing hundreds of

thousands of deaths yearly globally [1–5]. Therefore, newer strategies should be required in the development of newer therapeutic options to combat bacterial infection to address unmet patients' requirements as well as to mitigate the exponential increment in the financial burden on the global health sector. In this aspect, the synthesis of sustainable antibiotic materials might resolve issues associated with existing antibiotics.

Indeed, sustainable antibiotic materials, which are responsive towards stimuli like pH, moisture, temperature, and light, etc., offer an excellent alternative to traditional antibiotic materials. Such antibiotic materials are rapidly responsive to particular stimuli, thereby achieving more effectiveness in a short period of exposure and minimizing the

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effect of developing antibacterial resistance [6–9]. Numerous photo-responsive agents such as TiO_2 , ZnO , WS_2 , WO_3 , C_3N_4 , MXene, and Fe_2O_3 have been developed so far for the removal of environmental pollutants, including pharmaceutical compounds, dyes, and micro-organisms from water. However, relatively less photo-responsive activity and higher band gap values limit their applicability to end applications [10–13]. Therefore, newer photo-responsive agents need to be designed, which have narrow band gap values and enhance photo-responsive activity to kill/inhibit bacterial strains.

Bismuth (Bi) compounds are well known for the treatment of *Helicobacter pylori* infections in the stomach, which is related to ulcers. Bi can help in the synthesis of the cell wall and bind to the bacterial protein, eventually inhibiting the growth of the bacterial strain [14,15]. Bi-oxyhalide (BiOX , where $\text{X} = \text{Cl}$, I , and Br) considerably gained popularity among researchers because of their exceptional absorption in the visible region, and lower band gap value (Bi 6 s, O 2 p orbitals in valance and conduction band), thereby easily changes the band gap values by incorporating metals, polymers, and other surface functional groups [16–19]. Additionally, BiOX generated electron hole pairs, which react with oxygen and water molecules to produce reactive oxygen species (ROS), including hydroxyl radicals ($\bullet\text{OH}$), hydrogen peroxide (H_2O_2), and superoxide radicals ($\text{O}_2^\bullet$). The generation of ROS induced oxidative stress in bacterial strains, which led to membrane disruption and DNA damage, thereby cells death. The layered structure, optimal band gap value, and charge separation ability of BiOX makes exceptional candidate for photo-responsive biocidal agents [20–23]. Additionally, the incorporation of ZnO within the Bi-skeleton reduces the band gap again, which further allows these compounds to absorb a large intake of visible light and enhance their efficiency in bacterial degradation [24,25]. As Bi has less photo-responsive activity, researchers are looking towards the newer Bi-based materials by incorporating metals/non-metals to enhance photo-responsive activity.

Bee pollen (BP) is a sticky liquid secreted by bees and used for various diseases, as it contains several types of flavonoids, terpenoids, and other important organic compounds. BP has antibacterial and anti-inflammatory activity because of its ability to change the protein structure of bacteria, subsequently inhibiting/killing bacterial strains. Moreover, BP also acted as a reducing agent of the metals, which helped in the synthesis of nanoparticles (NPs) [26–28]. ZnO is a well-known and proven antimicrobial agent. Several studies have been published on the antibacterial effect of ZnO -NPs. They can easily generate ROS, which can help to damage the bacterial cell wall, leading to cell death. Moreover, Zn^{2+} ions can change protein structure by displacing the metal ions from the associated protein structure. Additionally, due to their positive charge, they can easily interact with the negatively charged cell membrane, leading to the lysis of the cell [29–33]. Combining the effect of three viz, Zn, Bi, and BP, and using them as photo-responsive biocidal agents is the next step towards stimuli-responsive antibiotics, which can sense particular stimuli and respond only in the presence of stimuli, otherwise remain ineffective. Moreover, Zn, Bi, and BP-based photo-responsive biocidal agents might be promising antibiotic materials mainly due to sustainable development and cost-effectiveness. Additional benefits associated with stimuli-responsive antibiotics are that they are very fast responsive, immediately start showing their effect when they come in contact with stimuli (photo-responsive). The unique combination of ZnO , Bi, and BP makes them exceptional candidates for photo-responsive biocidal agents.

The present study described the synthesis of bimetallic (Zn/Bi)-oxy-chloride-based nanosheets enmeshed rods nanocomposite by using bee propolis (ZBCBR) for photo-responsive biocidal agents. The synthesis of ZBCBR-based photo-responsive biocidal agents is sustainable, cost-effective, and simple co-precipitation. The as-prepared ZBCBR-based photo-responsive biocidal agents were characterized using several characterization techniques, such as scanning electron microscopy (SEM), energy-dispersive X-ray spectroscopy (EDX), X-ray diffraction

(XRD), Fourier-transform infrared spectroscopy (FTIR), diffuse reflectance spectroscopy (DRS), zeta potential (surface charge), and particle size. The as-prepared ZBCBR-based photo-responsive biocidal agents effectively kill/inhibit both gram-negative (*E. coli*) and gram-positive (*S. aureus*) bacterial strains. Moreover, ZBCBR-based photo-responsive biocidal agents show exceptional biocidal activity under solar irradiation in comparison with the dark. Therefore, the as-prepared ZBCBR-based photo-responsive biocidal agent is a facile and sustainable approach towards the development of photo-responsive biocidal agents.

2. Materials and methods

2.1. Chemicals

ZnSO_4 , Bi_2O_3 , hydrochloric acid (HCl), NaCl, and ethanol ($\text{C}_2\text{H}_5\text{OH}$) were procured from SRL chemicals, India. Yeast extract, tryptone, and agar were procured from Hi-Media Laboratory, India. BP was purchased from a local supplier in Mohali, Chandigarh, India. *E. coli* and *S. aureus* were obtained from a national research laboratory in India. All reagents used in this study were of the highest purity grade, and the solution was prepared using deionized (DI) water.

2.2. Synthesis of ZBCBR-based photo-responsive biocidal agent

The ZBCB-based photo-responsive biocidal agent was synthesized using a facile co-precipitation method. Briefly, 3.80 g of the Bi_2O_3 was dissolved in 25 mL DI and 2 mL of HCl solution with stirring for 60 min at 70 °C for 30 min. Next, 15 mL of DI was added into the solution with continued stirring at 70 °C to produce a BiOCl solution. ~1 g of ZnSO_4 was mixed into 25 mL of DI in a separate beaker and then added to the BiOCl solution, continuously stirring for 2 h at 70 °C to produce a homogenous solution of $\text{ZnSO}_4\text{-BiOCl}$. Approximately 0.5 g of BP was dissolved in 50 mL of DI under continuous stirring for 1 h at 50 °C to prepare the BP solution. The prepared BP solution was added drop-by-drop into a homogenous solution of $\text{ZnSO}_4\text{-BiOCl}$ with continuous stirring at 70 °C for 2 h, and then the temperature was switched off to produce ZBCB-nanosheets based intertwined rods. The synthesized ZBCBR-nanosheets based intertwined rods were washed several times with DI and $\text{C}_2\text{H}_5\text{OH}$. The washed ZBCBR-nanosheets based intertwined rod was dried at 70 °C for 24 h for further uses, named as ZBCBR-1. Likewise, without and with different amounts of metals, like 0.5 g and 2 g, were also prepared, namely ZBCBR-0, ZBCBR-0.5, and ZBCBR-2, respectively, for comparison purposes. Fig. 1. A schematic illustration of the synthesis and characterization of the ZBCBR-based photo-responsive biocidal agents.

2.3. Biocidal activity

The biocidal activity of the different ZBCBR-based photo-responsive agents was determined against both Gram-negative (*E. coli*) and Gram-positive (*S. aureus*) bacterial strains. Briefly, Luria-Bertani (LB) media (mixing ~10 g of tryptone, 5 g of yeast extract, and 10 g of NaCl in 1000 mL of deionized (DI) water) was prepared in a conical flask. Next, LB media was autoclaved at 121 °C for 20 min to sterilize. Approximately 200 μL of *E. coli* and *S. aureus* ($10^3\text{-}10^4$ CFU/mL) were inoculated in 20 mL of LB media, separately. Approximately 40 mg of different ZBCB-based photo-responsive biocidal agents were added to 20 mL of LB media with bacterial culture, in a separate conical flask. The LB media with bacterial culture served as a control. Next, conical flasks were incubated at 37 °C for 24 h. After a 24 h incubation period, the absorbance of each tube (growth of bacteria) was measured at 600 nm using a UV-Vis spectrophotometer (Shimadzu 2600).

2.3.1. Photo-responsive biocidal activity

The photo-responsive biocidal activity was ascertained by using a solar simulator (Xenon lamp, 35 W, 8000 K) against both Gram-negative

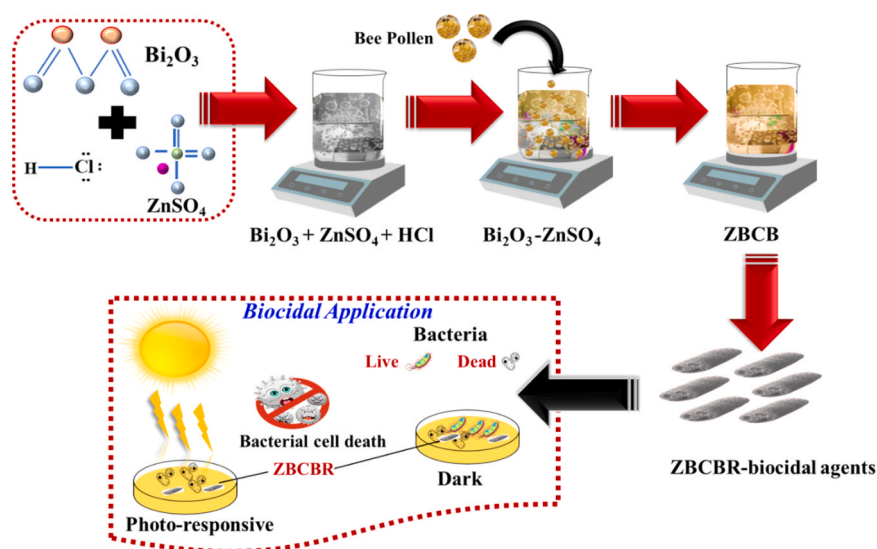


Fig. 1. A schematic illustration of the synthesis and characterization of the ZBCBR-based photo-responsive biocidal agents.

(*E. coli*) and Gram-positive (*S. aureus*) bacterial strains. Briefly, a glass photocatalytic reactor was filled with 200 mL of LB media, containing

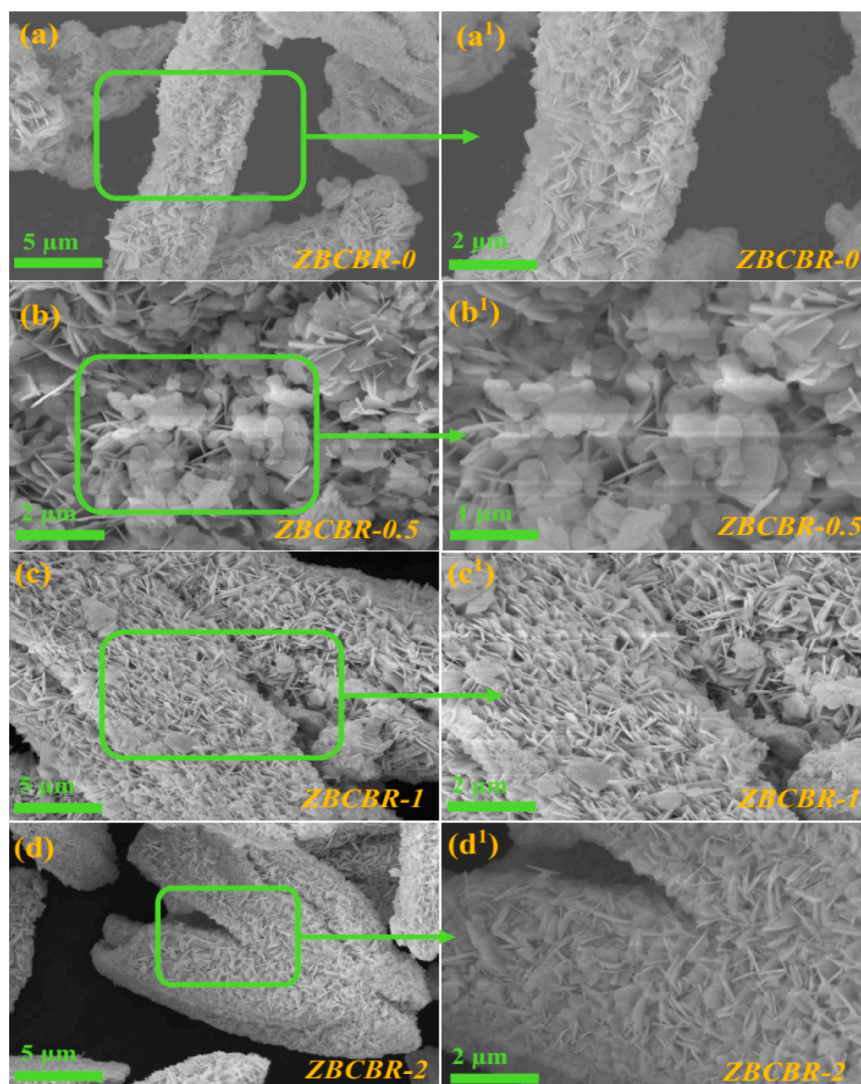


Fig. 2. SEM images of the different ZBCBR-based photo-responsive biocidal agents. (a-a¹). ZBCBR-0, (b-b¹). ZBCBR-0.5, (c-c¹). ZBCBR-1, and (d-d¹). ZBCBR-2.

E. coli and *S. aureus* (10^3 – 10^4 CFU/mL), separately. Next, 400 mg of different ZBCBR-based photo-responsive agents were added to LB media containing photocatalytic glass reactors. The photocatalytic reactor was kept in the dark for 30 min. Approximately ~ 1.5 mL of sample was taken at different time intervals (-30, 0, 5, 10, 20, 30, 40, 50, and 60 min). The collected treated samples were analyzed by measuring absorbance at 600 nm using a UV-Vis spectrophotometer. The bacteria control samples with light (without any ZBCBR-based photo-responsive biocidal agents) served as control. All experiments were performed in triplicate to check the reproducibility.

3. Material characterization

The as-prepared different ZBCBR-photo-responsive biocidal agents were characterized using several characterization techniques, including scanning electron microscopy (SEM), energy-dispersive X-ray spectroscopy (EDX), X-ray diffraction (XRD), diffuse reflectance spectroscopy (DRS), photoluminescence spectroscopy (PL), and Fourier transform infrared spectroscopy (FT-IR). The structural analysis of as-prepared ZBCBR photo-responsive biocidal agents was observed by SEM

analysis. The elemental analysis of as-prepared ZBCBR-photo-responsive biocidal agents was observed by EDX analysis (JEOL, JCM-7000). The diffraction patterns of the as-prepared ZBCB-photo-responsive biocidal agents were analyzed by using XRD analysis to understand the crystallographic indices with Cu K α radiation ($K\alpha = 1.541 \text{ \AA}$) (XR Dynamic 500 Diffractometer- Anton Paar). The band gap values of the different ZBCBR photo-responsive biocidal agents were calculated using the Tauc plot from DRS analysis. The recombination or separation of electron holes and surface defects in different ZBCBR photo-responsive biocidal agents was ascertained by using PL spectroscopic analysis (Shimadzu 6000 RF).

4. Results & discussion

4.1. Morphological and elemental analysis

Fig. 2 shows the SEM images of the different ZBCBR-based photo-responsive biocidal agents. Fig. 2(a-a¹) shows the SEM images of the ZBCBR-0 (BiOCl). The SEM images of ZBCBR-0 show nanosheets, which are enmeshed to produce a rod-like structure. The ZBCBR-0 shows the

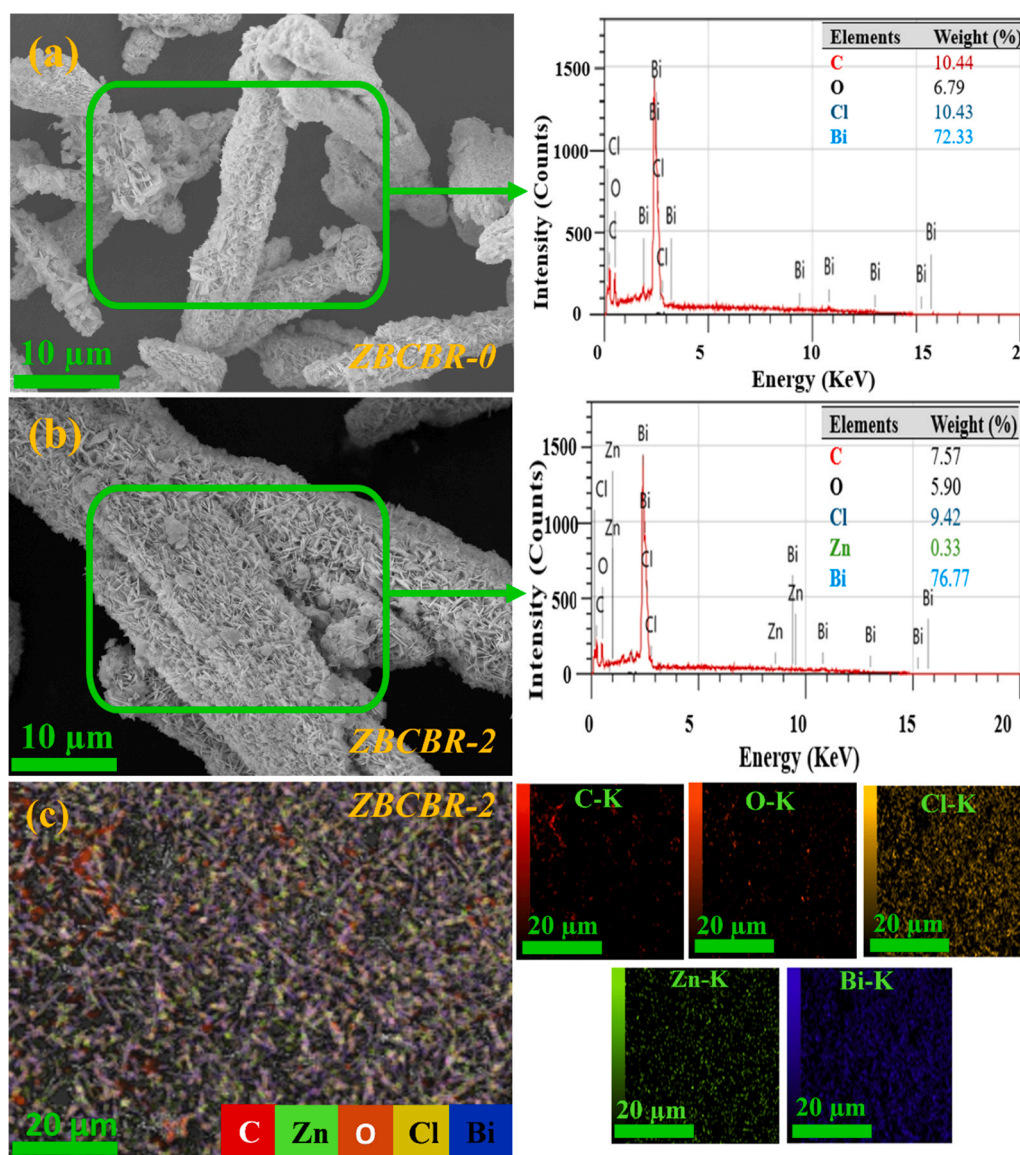


Fig. 3. (a-b) Elemental analysis of the ZBCBR-0 and ZBCBR-2-based photo-responsive biocidal agents, and (c) elemental mapping of the ZBCBR-2-based photo-responsive biocidal agents.

porous texture due to the nanosheet-based rod. Fig. 2(b-b¹) shows the SEM images of the ZBCBR-0.5. The SEM images show the nanosheet-like structure. Interestingly, no rod-shaped structure was observed upon incorporation of Zn-metals within BiOCl. As ZnO can break the assembled structures due to its high surface energy [34]. Fig. 2(c-c¹) shows the SEM images of the ZBCBR-1. The SEM images show the porous rod-like structure, which is assembled by the nanosheets. Interestingly, upon increasing the amount of Zn-metals from 0.5 g to 1 g, there is the formation of rod-shape-like structure is formed. Moreover, the intact structure of nanosheets and porosity might be advantageous for photo-responsive biocidal activity. Fig. 2(d-d¹) shows the SEM images of the ZBCBR-2. Like ZBCBR-1, a porous rod-shaped structure was observed. Moreover, the higher content of Zn-metals results in relatively less porosity and a less precise nanosheet structure due to agglomeration [35].

Fig. 3(a-c) shows the elemental analysis of the different CBZBR-based photo-responsive biocidal agents and mapping of CBZBR-2-based photo-responsive biocidal agents. The presence of C, Bi, O, and Cl attributed to the synthesis of BiOCl (Fig. 3(a)), whereas the presence of Zn, C, Bi, O, and Cl attributed to the successful synthesis of ZBCBR-based photo-responsive biocidal agents (Fig. 3(b)). Fig. 3(c) shows the elemental mapping of the different CBZBR-1-based photo-responsive biocidal agents. The elemental mapping indicates the even dispersal of the elements within the CBZBR-2, suggesting successful synthesis of the photo-responsive biocidal agents.

4.2. Structural and compositional analysis

Fig. 4(a) shows the XRD analysis of the synthesized different ZBCBR-based photo-responsive biocidal agents. Several prominent peaks were obtained in the spectra of all three samples. The peaks of ZBCBR-0, in which the most prominent peak appeared at 11.84° (001), 24.25° (101), 32.66° (110), 33.54° (102), and 35.01° (111) 2θ position. Other peaks were also observed at 40.81° (112), 50.21° (113), 58.25° (212), and 60.49° (203). All these peaks confirm the tetragonal structure of BiOCl [36,37]. The XRD spectra of ZBCBR-based photo-responsive biocidal agents, in which the prominent peak of Zn had also appeared, like 2θ position of 31.78°, 34.41°, 36.32°, 47.61°, and 56.63° correspond to (110), (002), (101), (102), and (110) planes. All peaks of the Zn are indexed in the Powder Diffraction Standards (JCPDS- 36-1451) [38,39]. As the peaks of Zn appeared almost at the same position as BiOCl [36]. There is a reduction in the intensity of the peak, and some peaks are merged and slightly shifted from the original position, showing the incorporation of Zn in the BiOCl skeleton. The emergence of a new peak at a slightly different position from the original, conforming to the synthesis of Zn-doped

BiOCl.

Fig. 4(b) demonstrates the FTIR spectra of different ZBCBR-based photo-responsive biocidal agents. Several prominent peaks are observed in the spectra, confirming the presence of metal-doped BiOCl. The most prominent peaks observed in the spectra were at 528, 890, 1458, 1754, 2978, and 3665 cm⁻¹. These peaks were observed in all samples, indicating metal-doped BiOCl. The first peak observed at 3665 cm⁻¹ (in all samples) belongs to the O-H stretching vibration due to the presence of inter- and intramolecular hydrogen bonding [40]. Another peak was observed at 2978 cm⁻¹, which is the C-H stretch, confirming the presence of aliphatic groups in the biopolymeric backbone, in the residual cellulose acetate and hydrocarbon chain [41]. The peak at the low frequency region of 890 cm⁻¹ corresponding to metal oxide M-O-H bending proves the presence of hydroxylated Zn species [42]. The most important peak confirming the formation of BiOCl appears at 528 cm⁻¹, which is due to the presence of the Bi-O bond. Again, the peak at 1458 confirms the presence of a Bi-Cl bond [43]. Furthermore, a strong peak at 1754 cm⁻¹ is identified only in ZBCBR-1 and ZBCBR-2, indicating the occurrence of H-O-H bending vibrations, which is due to the physical adsorption of water on the surface of the synthesized particle [44]. All these peaks in the spectra prove the successful synthesis of ZBCBR-based photo-responsive biocidal agents.

4.3. Particle size and Zeta potential

Fig. 5(a) shows the particle size of the different ZBCBR-based photo-responsive biocidal agents. The data indicate that upon the incorporation of Zn-metals within the ZBCBR, particle size significantly increased. Moreover, particle size significantly decreases upon further incorporation of Zn-metals from 0.5 g to 2 g. The particle size of the ZBCBR-0, ZBCBR-0.5, ZBCBR-1, and observed from the data, the largest particle size was observed to be ~296.06 nm of ZBCBR-0.5, which might make it difficult to form rods, due to their increasing steric hinderance and reduced flexibility, thereby making it difficult to form rods, which is also confirmed from SEM images.

Fig. 5(b) shows the surface of the different ZBCB-based photo-responsive biocidal agents. The negatively charged surface of ZBCBR-0 was observed to be ~ -3.767 eV. Upon incorporation of Zn-metals surface charge becomes positive. The surface charge of the ZBCBR-0.5, ZBCBR-1, and ZBCBR-2 was observed to be ~ -1.376 eV, ~13.66 eV, and ~28.27 eV. The positively charged surface might be beneficial for the antibacterial activity. The positively charged surface of the ZBCBR-based photo-responsive biocidal agents not only improved the antibacterial activity by electrostatic interactions with bacterial cells but also provides an advantage in photo-responsive biocidal activity by

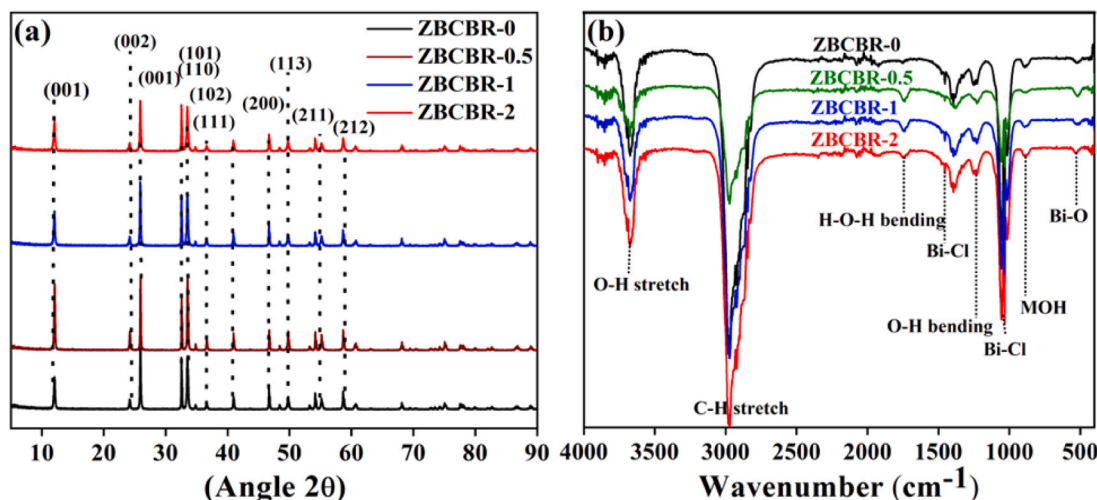


Fig. 4. (a). XRD and (b) FTIR of different ZBCBR-based photo-responsive biocidal agents.

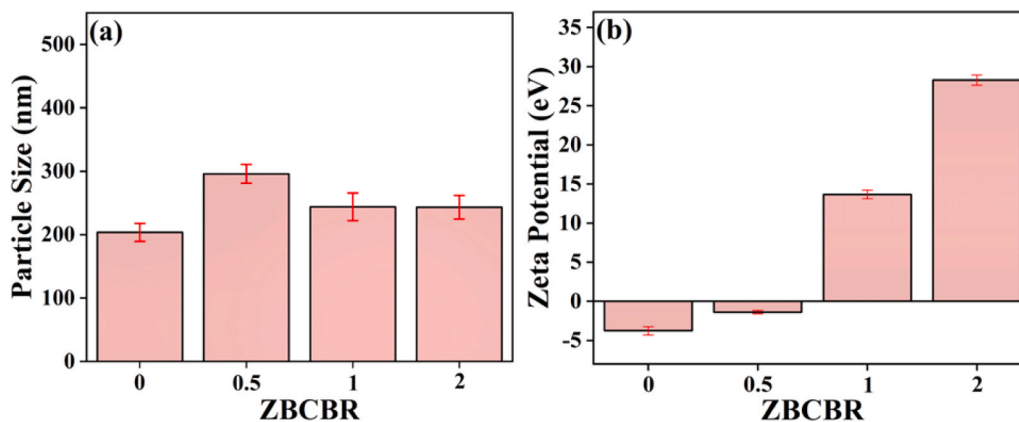


Fig. 5. (a) Particle size and (b) zeta potential of the ZBCBR-based photo-responsive biocidal agents. The data are presented with the mean ($n = 3$).

facilitating charge separation and production of ROS [45,46], which will be revisited in the subsequent section.

4.4. Band gap value

Fig. 6 shows the band gap value of the different ZBCBR-based photo-responsive biocidal agents. As observed from the data, the band gap values of the ZBCBR-0, ZBCBR-0.5, ZBCBR-1, and ZBCBR-2 were calculated to be ~ 2.65 eV, ~ 2.96 eV, ~ 2.58 eV, and ~ 2.42 eV, respectively. The incorporation of Zn-metals (0.5 g) within the ZBCBR significantly increased the band gap value, which might be due to the structural changes, which were earlier confirmed from SEM analysis. Although further incorporating Zn-metals from 0.5 g to 2 g, the band gap was significantly decreased, as there were changes in structural changes, which were earlier confirmed from the SEM analysis. The narrowing of the band gap value can be attributed to the successful incorporation of Zn-metals within the ZBCBR-based photo-responsive

biocidal agent. The non-monotonic variation in the band gap value upon doping Zn-metals within the ZBCBR-based photo-responsive biocidal agent, as at the lowest Zn-metal concentration, Zn^{2+} ions substitute host lattice cations, which leads to enhanced crystallinity and reduces defects, thereby widening the band gap value. On the other hand, at higher Zn-metal concentrations or excessive incorporation of dopant promotes the intrinsic defects like oxygen vacancies and surface defects, thereby narrowing the band gap value [47,48]. These findings are consistent with earlier studies where Zn-metal doping shows narrowing of the band gap values by incorporating different concentrations of Zn-metals [49–51]. The doping of Zn-metals within the ZBCBR-based photo-responsive biocidal agents introduces defects or oxygen vacancies, which can lead to the narrowing of the band gap or widening of the band gap value, which mainly depends on the dopants and their concentrations. The as-prepared ZBCBR-based photo-responsive biocidal agents observed narrowing band gap values, which enhance visible light absorption, thereby high photo-responsive biocidal activity.

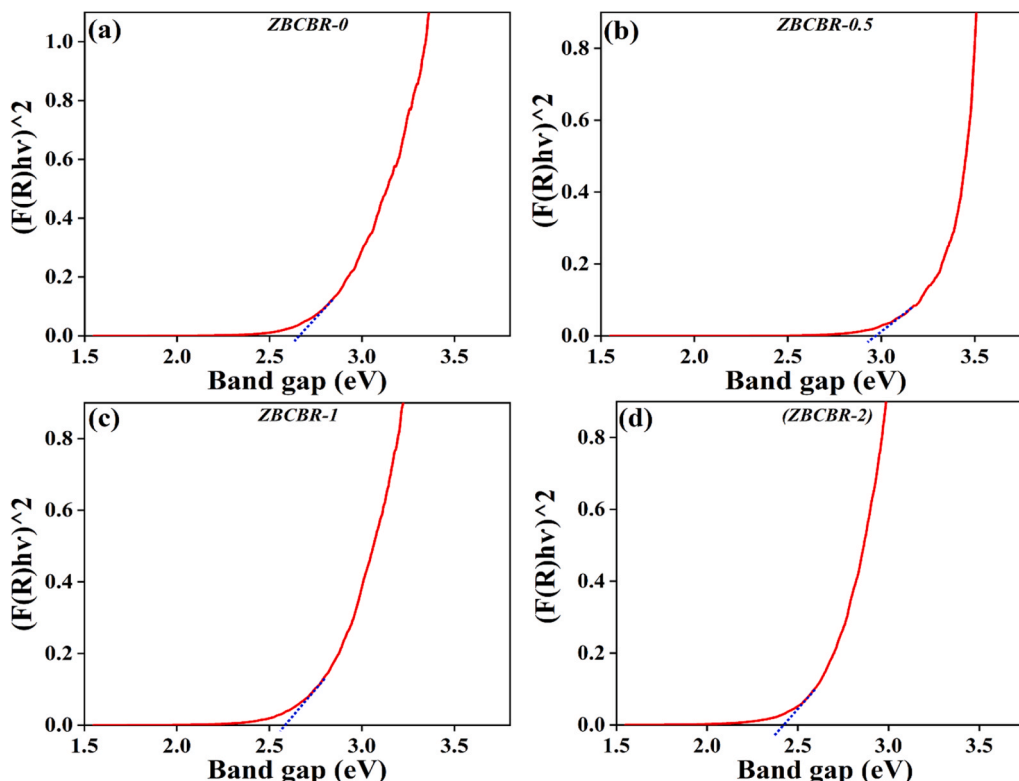


Fig. 6. Band gap of the different ZBCBR-based photo-responsive biocidal agents. (a). ZBCBR-0. (b). ZBCBR-0.5, (c). ZBCBR-1, and (d). ZBCBR-2.

4.5. PL spectroscopy

The PL spectroscopic analysis was performed to ascertain the surface defects and separation/recombination of electron holes. Fig. 7 shows the PL spectra of the different ZBCBR-based photo-responsive biocidal agents at a range of 450–620 nm by an excitation at 350 nm. As shown in the figure, significant changes were observed upon incorporation of Zn-metals within the ZBCBR-based photo-responsive biocidal agent. Initially, upon incorporation, 0.5 g of Zn-metals within the ZBCBR significantly increased the intensity, whereas upon increasing the amount of Zn-metals from 0.5 g to 2 g intensity significantly decreased with increasing the amount of Zn-metals. The data were consistent with the results of SEM and particle size, as discussed earlier in the manuscript. It is important to note that all ZBCBR photo-responsive show the two characteristic peaks at 470 nm and 541 nm. The peak at 470 nm is attributed to the recombination of free excitation, whereas the peak at 541 nm is attributed to defects, mainly oxygen defects. Moreover, upon the incorporation of Zn-metals within the ZBCBR, there were slight changes in the peaks observed. The incorporation of Zn-metals within the ZBCBR significantly decreases the intensity of PL spectra, which contributed to enhanced separation of charge carriers. Another aspect, the incorporation of Zn within the ZBCBR-based photo-responsive biocidal agent, induces luminescence quenching by altering the lattice structure or structural changes. The quenching effect is mainly due to the formation of defect states and oxygen vacancies [52,53], thereby suppressing the recombination of electron-holes, subsequently improving photo-responsive biocidal activity.

4.6. Biocidal performance

Fig. 8 shows the biocidal activity of the different ZBCBR-based photo-responsive biocidal agents. Fig. 8a shows the biocidal activity of different ZBCBR-based photo-responsive biocidal agents (2 mg/mL) against *E. coli* for 24 h of exposure. As observed from the figure, approximately ~76% of bacterial cells were killed/inhibited within 24 h of exposure. Moreover, upon incorporation of Zn-metals within the ZBCBR, bacterial inhibition significantly increased. The antibacterial activity significantly increased with increasing Zn-metals. The ZBCBR-0.5, ZBCBR-1, and ZBCBR-2 shows bacterial cell viability of ~ 85%, ~ 93%, and ~ 95%, respectively.

Fig. 8b shows the biocidal activity of different ZBCBR-based photo-responsive biocidal agents (2 mg/mL) against *S. aureus* for 24 h of exposure. As observed from the figure, approximately 76%, 77%, 89%, and 88% bacterial inhibition of ZBCBR-0, ZBCBR-0.5, ZBCBR-1, and ZBCBR-2, respectively. The data was consistent with *E. coli*, showing that increasing the Zn-metals significantly enhanced antibacterial activity. The ZBCBR-based photo-responsive biocidal agents disrupt bacterial cells, producing ROS, thereby demonstrating antibacterial activity [54–56]. Based on the data, ZBCBR-2-based photo-responsive biocidal agents have greater biocidal activity among all synthesized

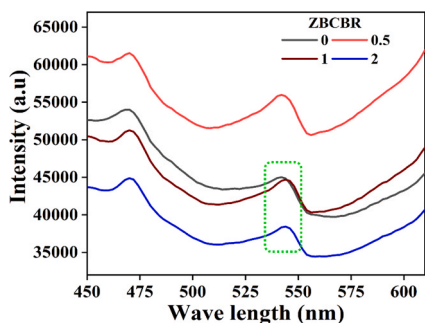


Fig. 7. PL emission spectra at 350 nm excitation of the different ZBCBR-based photo-responsive biocidal agents.

ZBCBR-based photo-responsive biocidal agents, which were further analyzed at different time intervals and for photo-responsive biocidal activity.

4.6.1. Photo-responsive-biocidal activity

Fig. 9(a–b) shows the photo-responsive biocidal activity of the ZBCBR-2 against *E. coli* and *S. aureus* under solar irradiation. The bacteria control samples with light (without any ZBCBR-based photo-responsive biocidal agents) show no significant bacterial inhibition, suggesting antibacterial activity mainly due to photo-responsive biocidal agents (data were not presented due to brevity). As observed from the figure, ~99.99% bacterial cell death was observed within 5 min of solar irradiation at a 2 mg/mL dose, whereas ~99.99% bacterial cells were killed within 10 min at 1 mg/mL against both *E. coli* and *S. aureus*. Interestingly, the data suggested that the as-prepared ZBCBR photo-responsive biocidal agents have exceptional biocidal activity under solar light irradiation compared with that in the dark.

The photo-responsive biocidal activity is mainly due to the synergistic interaction between Zn-metal and BiOCl. As BiOCl exhibits photo-responsive characteristics by producing ROS, mainly superoxide ($O_2^{\bullet-}$), and hydroxyl radicals ($\bullet OH$). The ROS are highly reactive and able to disrupt the cellular membrane of bacteria, subsequently cell death. Fig. 10 shows the plausible mechanism of photo-responsive biocidal agents against bacterial strains. The incorporation of Zn-metals within the ZBCBR significantly narrows the band gap values, which enhances the separation and transfer of photogenerated charge carriers, subsequently lowering the electron-hole recombination rate. Furthermore, Zn-metals themselves contribute biocidal activity through the release of Zn^{2+} ions, which interfere with the metabolism of bacteria by inducing oxidative stress [55,57–59]. The combined effect of Zn and BiOCl within the ZBCBR significantly enhances photo-responsive biocidal activity. Therefore, the prepared ZBCBR-based photo-responsive biocidal agents effective against bacterial strains within 5 min of solar irradiation at a 2 mg/mL dose.

5. Conclusion

The ZBCBR-based photo-responsive biocidal agent was synthesized using a simple sol-gel process using green chemistry. The maximum ~95% biocidal activity was observed at 24 h for ZBCBR-2 photo-responsive biocidal agent in the dark, whereas ~99.99% bacterial cells were killed within 5 min of solar irradiation at 2 mg/mL. The outstanding photo-responsiveness of ZBCBR-2 against both *E. coli* and *S. aureus* is attributed to its narrow band gap, which has been knowingly improved by oxygen vacancies. Furthermore, Zn-metals increase the surface defects and reduce the recombination rate, thereby enhancing photo-responsive biocidal activity. Moreover, ZBCBR-based photo-responsive biocidal agents are promising antibiotic materials due to sustainability and cost-effectiveness. The scalability, long-term stability, environmental safety, and cytotoxicity remain unexplored. Indeed, large-scale production must ensure consistent quality and photo-responsive biocidal activity, whereas long-term exposure might lead to the metal ions leaching, and degradation of ZBCBR-based photo-responsive biocidal agent, which are essential for their safe and real-time applications. Therefore, the synthesized ZBCBR-based photo-responsive biocidal agent is a cost-effective and sustainable approach towards the development of a photo-responsive biocidal agent to control bacterial infection.

CRediT authorship contribution statement

Mohammad Ashfaq: Writing – review & editing, Supervision, Formal analysis, Conceptualization. **Werayut Srituravanich:** Writing – review & editing. **Divya Chauhan:** Writing – review & editing, Formal analysis. **Sangeeta Sahu:** Writing – review & editing. **Shruti Devi:** Writing – original draft, Methodology, Investigation. **Neetu Talreja:**

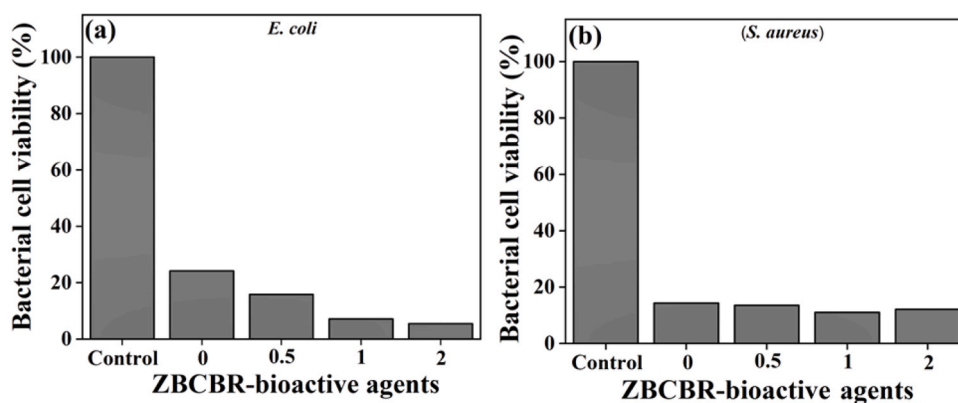


Fig. 8. Biocidal activity of the different ZBCBR-based photo-responsive biocidal agents. (a) *E. coli*, and (b) *S. aureus*. The data are presented with the mean ($n = 3$).

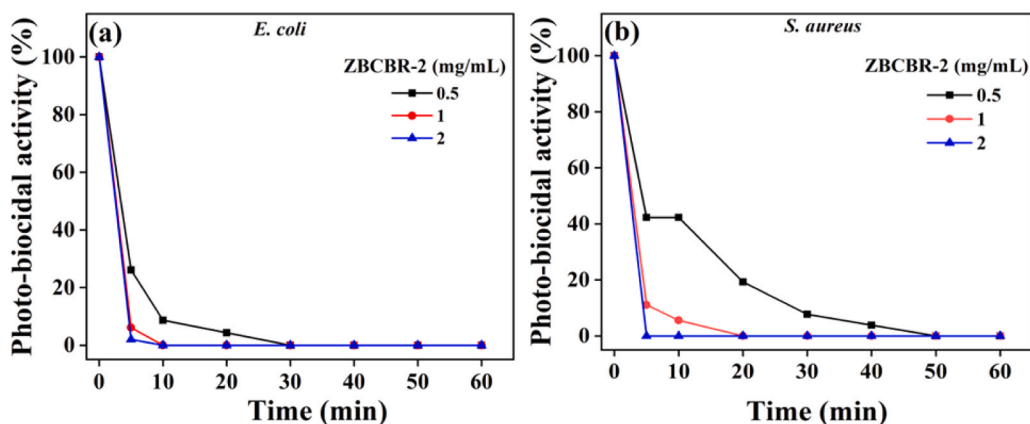


Fig. 9. Photo-responsive biocidal activity of ZBCBR-2. (a) *E. coli*, and (b) *S. aureus*. The data are presented with the mean ($n = 3$).

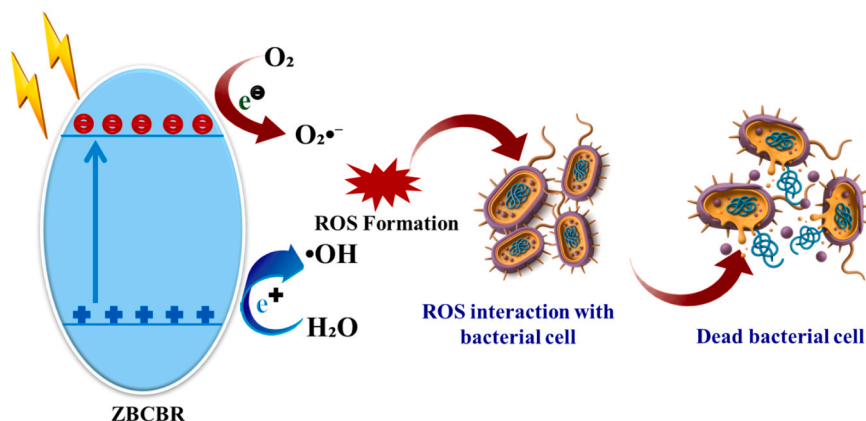


Fig. 10. A plausible mechanism of photo-responsive biocidal agents against bacterial strains.

Writing – review & editing, Investigation, Formal analysis, Conceptualization. Archsina Maimom: Writing – original draft, Methodology, Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] T.M. Uddin, A.J. Chakraborty, A. Khuro, B.M.R.M. Zidan, S. Mitra, T.B. Emran, K. Dhama, M.K.H. Ripon, M. Gajdacs, M.U.K. Sahibzada, M.J. Hossain, N. Koirala, Antibiotic resistance in microbes: history, mechanisms, therapeutic strategies and future prospects, *J. Infect. Public Health* 14 (12) (2021) 1750–1766, <https://doi.org/10.1016/j.jiph.2021.10.020>.

- [2] M.A. Salam, M.Y. Al-Amin, M.T. Salam, J.S. Pawar, N. Akhter, A.A. Rabaan, M.A. Alqumber, Antimicrobial resistance: a growing serious threat for global public health, *Healthcare* 11 (13) (2023), <https://doi.org/10.3390/healthcare11131946>.
- [3] G. Muteeb, M.T. Rehman, M. Shahwan, M. Aatif, Origin of antibiotics and antibiotic resistance, and their impacts on drug development: a narrative review, *Pharmaceuticals* 16 (11) (2023), <https://doi.org/10.3390/ph16111615>.
- [4] S.K. Ahmed, S. Hussein, K. Qurbani, R.H. Ibrahim, A. Fareeq, K.A. Mahmood, M. G. Mohamed, Antimicrobial resistance: impacts, challenges, and future prospects, *J. Med. Surg. Public Health* 2 (2024) 100081, <https://doi.org/10.1016/j.glmedi.2024.100081>.
- [5] D.G.J. Larsson, C.-F. Flach, Antibiotic resistance in the environment, *Nat. Rev. Microbiol.* 20 (5) (2022) 257–269, <https://doi.org/10.1038/s41579-021-00649-x>.
- [6] X. Wang, M. Shan, S. Zhang, X. Chen, W. Liu, J. Chen, X. Liu, Stimuli-responsive antibacterial materials: molecular structures, design principles, and biomedical applications, *Adv. Sci.* 9 (13) (2022) e2104843, <https://doi.org/10.1002/adv.202104843>.
- [7] J. Zhang, W. Tang, X. Zhang, Z. Song, T. Tong, An overview of stimuli-responsive intelligent antibacterial nanomaterials, *Pharmaceutics* 15 (8) (2023), <https://doi.org/10.3390/pharmaceutics15082113>.
- [8] Y. Huang, X. Guo, Y. Wu, X. Chen, L. Feng, N. Xie, G. Shen, Nanotechnology's frontier in combatting infectious and inflammatory diseases: prevention and treatment, *Signal Transduct. Target. Ther.* 9 (1) (2024) 34, <https://doi.org/10.1038/s41392-024-01745-z>.
- [9] C. Wang, P. Xu, X. Li, Y. Zheng, Z. Song, Research progress of stimulus-responsive antibacterial materials for bone infection, *Front. Bioeng. Biotechnol.* 10 (2022) 2022, <https://doi.org/10.3389/fbioe.2022.1069932>.
- [10] H. Kumari, Sonia, R. Suman, S. Ranga, S. Chahal, S. Devi, S. Sharma, P. Kumar, S. Kumar, A. Kumar, R. Kumar, Parmar, A review on photocatalysis used for wastewater treatment: dye degradation, *Water Air Soil Pollut.* 234 (6) (2023) 349, <https://doi.org/10.1007/s11270-023-06359-9>.
- [11] R. Ghamarpoor, A. Fallah, M. Jamshidi, A review of synthesis methods, modifications, and mechanisms of ZnO/TiO₂-based photocatalysts for photodegradation of contaminants, *ACS Omega* 9 (24) (2024) 25457–25492, <https://doi.org/10.1021/acsomega.3c08717>.
- [12] N. Gaur, D. Dutta, A. Singh, R. Dubey, D.V. Kamboj, Recent advances in the elimination of persistent organic pollutants by photocatalysis, *Front. Environ. Sci.* 10 (2022) 2022, <https://doi.org/10.3389/fenvs.2022.872514>.
- [13] M. Khodamorady, K. Bahrami, Fe(3O)(4)@BNPs@ZnO-ZnS as a novel, reusable and efficient photocatalyst for dye removal from synthetic and textile wastewaters, *Heliyon* 9 (6) (2023) e16397, <https://doi.org/10.1016/j.heliyon.2023.e16397>.
- [14] H. Alkim, A.R. Koksai, S. Boga, I. Sen, C. Alkim, Role of Bismuth in the Eradication of *Helicobacter pylori*, *Am. J. Ther.* 24 (6) (2017) e751–e757, <https://doi.org/10.1097/mjt.0000000000000389>.
- [15] J.D.S. Rosário, F.H. Moreira, L.H.F. Rosa, W. Guerra, P.P. Silva-Caldeira, Biological Activities of Bismuth Compounds: An Overview of the New Findings and the Old Challenges Not Yet Overcome, *Molecules* 28 (15) (2023), <https://doi.org/10.3390/molecules28155921>.
- [16] T.L. Yusuf, B.O. Orimolade, D. Masekela, K.A. Adegoke, K.D. Modibane, S. Makgato, BIO(X)=Cl, Br, I)-based S-scheme heterostructure photocatalysts for environmental remediation and energy conversion, *Mater. Today Sustain.* 30 (2025) 101115, <https://doi.org/10.1016/j.mtsust.2025.101115>.
- [17] Z. Xiang, X. Wang, X. Jiang, C. Li, Y. Ding, Q. Zheng, L. Zhou, H. Zhu, Q. Wang, Y. Wu, L. He, G. Yang, Bismuth oxyhalides (BiOX (X = Cl, Br, I)) based composites for photocatalytic antibacterial: Modification strategies, antibacterial mechanisms and prospects, *J. Environ. Chem. Eng.* 13 (3) (2025) 116428, <https://doi.org/10.1016/j.jece.2025.116428>.
- [18] X. Wei, M.U. Akbar, A. Raza, G. Li, A review on bismuth oxyhalide based materials for photocatalysis, *Nanoscale Adv.* 3 (12) (2021) 3353–3372, <https://doi.org/10.1039/D1NA00223F>.
- [19] G.X. Castillo-Cabrera, P.J. Espinoza-Montero, P. Alulema-Pullupaxi, J.R. Mora, M. H. Villacís-García, Bismuth Oxyhalide-Based Materials (BiOX: X = Cl, Br, I) and Their Application in Photoelectrocatalytic Degradation of Organic Pollutants in Water: A Review, *Front. Chem.* 10 (2022) 2022, <https://doi.org/10.3389/fchem.2022.900622>.
- [20] X. Sun, Y. Zhang, C. Li, Z. Zhang, Z. Peng, H. Si, J. Zhang, Y. Li, BiOClxBr_{1-x} (x+y+z=1) solid solutions with controllable band gap and highly enhanced visible light photocatalytic performances, *J. Alloy. Compd.* 638 (2015) 254–260, <https://doi.org/10.1016/j.jallcom.2015.03.150>.
- [21] Y. Sheng, I.A. Abreu, D.E. Cabelli, M.J. Maroney, A.-F. Miller, M. Teixeira, J. S. Valentine, Superoxide Dismutases and Superoxide Reductases, *Chem. Rev.* 114 (7) (2014) 3854–3918, <https://doi.org/10.1021/cr4005296>.
- [22] K.M. Sutherland, A. Coe, R.J. Gast, S. Plummer, C.P. Suffridge, J.M. Diaz, J. S. Bowman, S.D. Wankel, C.M. Hansel, Extracellular superoxide production by key microbes in the global ocean, *Limnol. Oceanogr.* 64 (6) (2019) 2679–2693, <https://doi.org/10.1002/lno.11247>.
- [23] N. Talreja, D. Chauhan, J. Sindya, E. Perumal, M. Fareed, M. Ashfaq, Synthesis of bimetallic (Fe/Bi) oxychloride nanosheet-entangled rods (FBOCNR) for osteosarcoma nano-therapy, *Mater. Chem. Phys.* 348 (2026) 131580, <https://doi.org/10.1016/j.matchemphys.2025.131580>.
- [24] M.A. Rahman, M.T. Hossain, M.F. Ahmed, M. Shahriar Bashar, S.S. Dey, S. Ahmed, M. Sahadat Hossain, Tuning the antimicrobial and photocatalytic activity of nano-ZnO by metal doping, *Mater. Adv.* 6 (11) (2025) 3686–3704, <https://doi.org/10.1039/D5MA00101C>.
- [25] Y. Li, C. Liao, S.C. Tjong, Recent Advances in Zinc Oxide Nanostructures with Antimicrobial Activities, *Int J. Mol. Sci.* 21 (22) (2020), <https://doi.org/10.3390/ijms21228836>.
- [26] F. Giampieri, J.L. Quiles, D. Ciansiosi, T.Y. Forbes-Hernández, F.J. Orantes-Bermejo, J.M. Alvarez-Suarez, M. Battino, Bee Products: An Emblematic Example of Underutilized Sources of Bioactive Compounds, *J. Agric. Food Chem.* 70 (23) (2022) 6833–6848, <https://doi.org/10.1021/acs.jafc.1c05822>.
- [27] C. Rodríguez-Pólit, R. Gonzalez-Pastor, J. Heredia-Moya, S.E. Carrera-Pacheco, F. Castillo-Solis, R. Vallejo-Imbaquingo, C. Barba-Ostria, L.P. Guamán, Chemical Properties and Biological Activity of Bee Pollen, *Molecules* 28 (23) (2023), <https://doi.org/10.3390/molecules28237768>.
- [28] A. Rzepecka-Stojko, J. Stojko, A. Kurek-Górecka, M. Górecki, A. Kabala-Dzik, R. Kubina, A. Moździerz, E. Buszman, Polyphenols from Bee Pollen: Structure, Absorption, Metabolism and Biological Activity, *Molecules* 20 (12) (2015) 21732–21749, <https://doi.org/10.3390/molecules201219800>.
- [29] A. Joe, S.-H. Park, K.-D. Shim, D.-J. Kim, K.-H. Jhee, H.-W. Lee, C.-H. Heo, H.-M. Kim, E.-S. Jang, Antibacterial mechanism of ZnO nanoparticles under dark conditions, *J. Ind. Eng. Chem.* 45 (2017) 430–439, <https://doi.org/10.1016/j.jiec.2016.10.013>.
- [30] N. Babayevska, I. Przysiecka, G. Iatsunskyi, M. Nowaczyk, E. Jarek, S. Janiszewska, Jurga, ZnO size and shape effect on antibacterial activity and cytotoxicity profile, *Sci. Rep.* 12 (1) (2022) 8148, <https://doi.org/10.1038/s41598-022-12134-3>.
- [31] H. Yang, J. Zhang, Z. Li, J. Huang, J. Wu, Y. Zhang, H. Ge, Y. Zhao, Antibacterial Effect of Low-Concentration ZnO Nanoparticles on Sulfate-Reducing Bacteria under Visible Light, *Nanomater. (Basel)* 13 (14) (2023), <https://doi.org/10.3390/nano13142033>.
- [32] B. Qu, Z. Xiao, Y. Luo, Sustainable nanotechnology for food preservation: Synthesis, mechanisms, and applications of zinc oxide nanoparticles, *J. Agric. Food Res.* 19 (2025) 101743, <https://doi.org/10.1016/j.jafr.2025.101743>.
- [33] M.A. Rauf, M. Owais, R. Rajpoot, F. Ahmad, N. Khan, S. Zubair, Biomimetically synthesized ZnO nanoparticles attain potent antibacterial activity against less susceptible *S. aureus* skin infection in experimental animals, *RSC Adv.* 7 (58) (2017) 36361–36373, <https://doi.org/10.1039/C7RA05040B>.
- [34] A. Al-Rasheedi, N.H. Alonizan, M.S. Aida, Effect of Zinc Nitrate Concentration on the Morphology of ZnO Nanostructures Grown by Hydrothermal Method, *Phys. Status Solidi (a)* 220 (12) (2023) 2200905, <https://doi.org/10.1002/pssa.202200905>.
- [35] S. Raha, M. Ahmaruzzaman, ZnO nanostructured materials and their potential applications: progress, challenges and perspectives, *Nanoscale Adv.* 4 (8) (2022) 1868–1925, <https://doi.org/10.1039/D1NA00880C>.
- [36] Y. Xu, Z. Shi, Le Zhang, E.M.B. Brown, A. Wu, Layered bismuth oxyhalide nanomaterials for highly efficient tumor photodynamic therapy, *Nanoscale* 8 (25) (2016) 12715–12722, <https://doi.org/10.1039/C5NR04540A>.
- [37] W. Ashraf, S. Bansal, V. Singh, S. Barman, M. Khanuja, BiOCl/WS₂ hybrid nanosheet (2D/2D) heterojunctions for visible-light-driven photocatalytic degradation of organic/inorganic water pollutants, *RSC Adv.* 10 (42) (2020) 25073–25088, <https://doi.org/10.1039/D0RA02916E>.
- [38] E.P. Etape, J. Foba-Tendo, L.J. Ngolui, B.V. Namondo, F.C. Yollande, M.B. N. Nguemong, Structural Characterization and Magnetic Properties of Undoped and Ti-Doped ZnO Nanoparticles Prepared by Modified Oxalate Route, *J. Nanomater.* 2018 (1) (2018) 9072325, <https://doi.org/10.1155/2018/9072325>.
- [39] O.A. Zelekew, S.G. Aragaw, F.K. Sabir, D.M. Andoshe, A.D. Duma, D.-H. Kuo, X. Chen, T.D. Desissa, B.B. Tesfamariam, G.B. Feyisa, H. Abdullah, E.T. Bekele, F. G. Aga, Green synthesis of Co-doped ZnO via the accumulation of cobalt ion onto *Eichhornia crassipes* plant tissue and the photocatalytic degradation efficiency under visible light, *Mater. Res. Express* 8 (2) (2021) 025010, <https://doi.org/10.1088/2053-1591/abe2d6>.
- [40] R.M. Silverstein, G.C. Bassler, Spectrometric identification of organic compounds, *J. Chem. Educ.* 39 (11) (1962) 546.
- [41] R. Deepa, D. Muthuraj, E. Kumar, V. Veeraputhiran, Microwave assisted synthesis of zinc oxide nanoparticles and its antimicrobial efficiency, *J. Nanosci. Technol.* (2019) 640–641.
- [42] N. Padmavathy, R. Vijayaraghavan, Enhanced bioactivity of ZnO nanoparticles—an antimicrobial study, *Sci. Technol. Adv. Mater.* (2008).
- [43] Z.S. Seddigi, M.A. Gondal, U. Baig, S.A. Ahmed, M.A. Abdulaziz, E.Y. Danish, M. M. Khaled, A. Lais, Facile synthesis of light harvesting semiconductor bismuth oxychloride nano photo-catalysts for efficient removal of hazardous organic pollutants, *PLOS ONE* 12 (2) (2017) e0172218, <https://doi.org/10.1371/journal.pone.0172218>.
- [44] L. Baia, R. Stefan, W. Kiefer, J. Popp, S. Simon, Structural investigations of copper doped BiO₃-Bi₂O₃ glasses with high bismuth oxide content, *J. Non-Cryst. Solids* 303 (3) (2002) 379–386.
- [45] Q. Su, Y. Xue, C. Wang, Q. Zhou, Y. Zhao, J. Su, B. Zhu, Strategies and applications of antibacterial surface-modified biomaterials, *Bioact. Mater.* 53 (2025) 114–140, <https://doi.org/10.1016/j.bioactmat.2025.07.009>.
- [46] H.T. Draviana, I. Fitriannisa, A. Jazidie, D.I. Krisnawati, M. Khafid, T.-R. Kuo, Antibacterial Mechanisms of Negatively and Positively Charged Ligands on Gold Nanoclusters, *ACS Appl. Nano Mater.* 8 (13) (2025) 6380–6390, <https://doi.org/10.1021/acsanm.4c07269>.
- [47] Y. Tang, R. Tai, X. Song, S. Gao, R. Wu, S. Chen, P. Li, Q. Wang, Zinc-Doped BiOBr Hollow Microspheres for Enhanced Visible Light Photocatalytic Degradation of Antibiotic Residues, *Langmuir* 40 (12) (2024) 6515–6523, <https://doi.org/10.1021/acs.langmuir.4c00155>.
- [48] C. Guan, T. Hou, W. Nie, Q. Zhang, L. Duan, X. Zhao, Enhanced photocatalytic reduction of CO₂ on BiOBr under synergistic effect of Zn doping and induced oxygen vacancy generation, *J. Colloid Interface Sci.* 633 (2023) 177–188, <https://doi.org/10.1016/j.jcis.2022.11.106>.

- [49] N. Kamarulzaman, M.F. Kasim, R. Rusdi, Band Gap Narrowing and Widening of ZnO Nanostructures and Doped Materials, *Nanoscale Res Lett.* 10 (1) (2015) 1034, <https://doi.org/10.1186/s11671-015-1034-9>.
- [50] S.A. Ansari, M.M. Khan, S. Kalathil, A. Nisar, J. Lee, M.H. Cho, Oxygen vacancy induced band gap narrowing of ZnO nanostructures by an electrochemically active biofilm, *Nanoscale* 5 (19) (2013) 9238–9246, <https://doi.org/10.1039/C3NR02678G>.
- [51] S. Guan, Y. Cheng, L. Hao, H. Yoshida, C. Tarashima, T. Zhan, T. Itoi, T. Qiu, Y. Lu, Oxygen vacancies induced band gap narrowing for efficient visible-light response in carbon-doped TiO₂, *Sci. Rep.* 13 (1) (2023) 14105, <https://doi.org/10.1038/s41598-023-39523-6>.
- [52] F.-M. Chang, S. Brahma, J.-H. Huang, Z.-Z. Wu, K.-Y. Lo, Strong correlation between optical properties and mechanism in deficiency of normalized self-assembly ZnO nanorods, *Sci. Rep.* 9 (1) (2019) 905, <https://doi.org/10.1038/s41598-018-37601-8>.
- [53] T. Munawar, S. Fatima, K.M. Batoo, M.S. Nadeem, F. Mukhtar, S. Hussain, S. A. Khan, M. Koc, F. Iqbal, Enhanced charge carriers separation/transportation via S-scheme ZnCeS–ZnWO heterostructure nanocomposite for photodegradation of synthetic dyes under sunlight, *Mater. Chem. Phys.* 314 (2024) 128938, <https://doi.org/10.1016/j.matchemphys.2024.128938>.
- [54] B. Lallo da Silva, M.P. Abuçafy, E. Berbel Manaia, J.A. Oshiro Junior, B.G. Chiari-Andréo, R.C.R. Pietro, L.A. Chiavacci, Relationship between structure and antimicrobial activity of zinc oxide nanoparticles: an overview, *Int J. Nanomed.* 14 (2019) 9395–9410, <https://doi.org/10.2147/ijn.S216204>.
- [55] S. Alfei, G.C. Schito, A.M. Schito, G. Zuccari, Reactive oxygen species (ROS)-mediated antibacterial oxidative therapies: available methods to generate ros and a novel option proposal, *Int J. Mol. Sci.* 25 (13) (2024), <https://doi.org/10.3390/ijms25137182>.
- [56] S. Hosseinzadeh-Moghaddam, M. Ashfaq, H. Ehtesabi, Synthesis of rice husk-zinc oxide-chitosan-based sponge: a green and eco-friendly nanocomposite for enhanced antibacterial, hemostasis, and wound healing applications, *J. Environ. Chem. Eng.* 12 (6) (2024) 114705, <https://doi.org/10.1016/j.jece.2024.114705>.
- [57] L. Fialho, A. Costa-Barbosa, P. Sampaio, S. Carvalho, Effects of Zn–ZnO core-shell nanoparticles on antimicrobial mechanisms and immune cell activation, *ACS Appl. Nano Mater.* 6 (18) (2023) 17149–17160, <https://doi.org/10.1021/acsanm.3c03241>.
- [58] S. Ragini, N. Devi, D. Talreja, S. Chauhan, M. Sahu, Ashfaq, Bioresource-derived Propolis wrapping for ZnO/TiO₂ nanocomposites: a sustainable photo-antibiotic material, *Inorg. Chem. Commun.* 186 (2026) 116270, <https://doi.org/10.1016/j.inoche.2026.116270>.
- [59] S. Mustafa, H.M. Khan, I. Shukla, F. Shujatullah, M. Shahid, M. Ashfaq, A. Azam, Effect of ZnO nanoparticles on ESBL producing *Escherichia coli* & *Klebsiella spp.*, *East. J. Med.* 16 (4) (2011) 253–257.